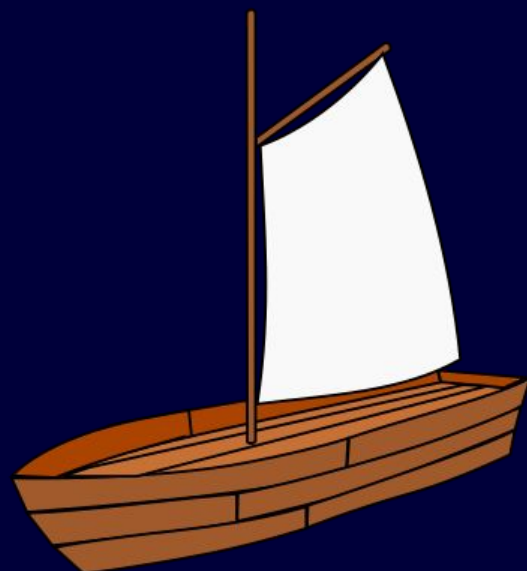
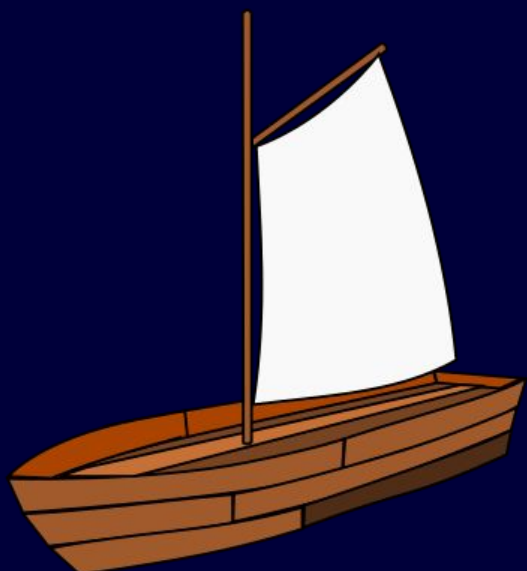
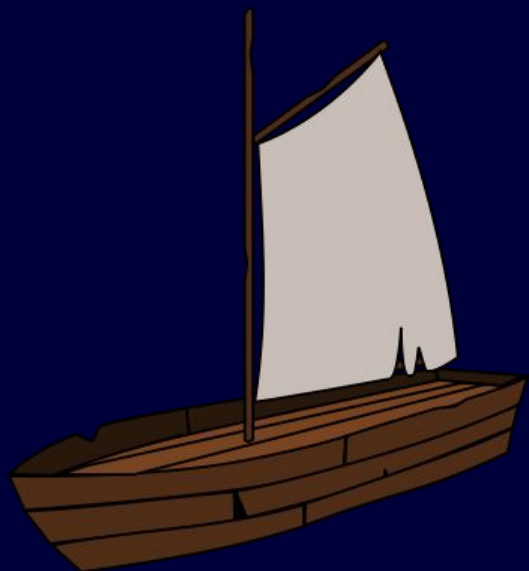


Automated Testing of a Large Language Model: A Live Demo

Anupam Krishnamurthy

Test Automation Days 2026

5.3.2026



Anupam Krishnamurthy

- ❖ MBA turned engineer
- ❖ Independent Test Automation Consultant
- ❖ Interests: Chess, Running, History

My LinkedIn



Challenges with LLM Testing

- ❖ Non-determinism
- ❖ Cost
- ❖ Architecture

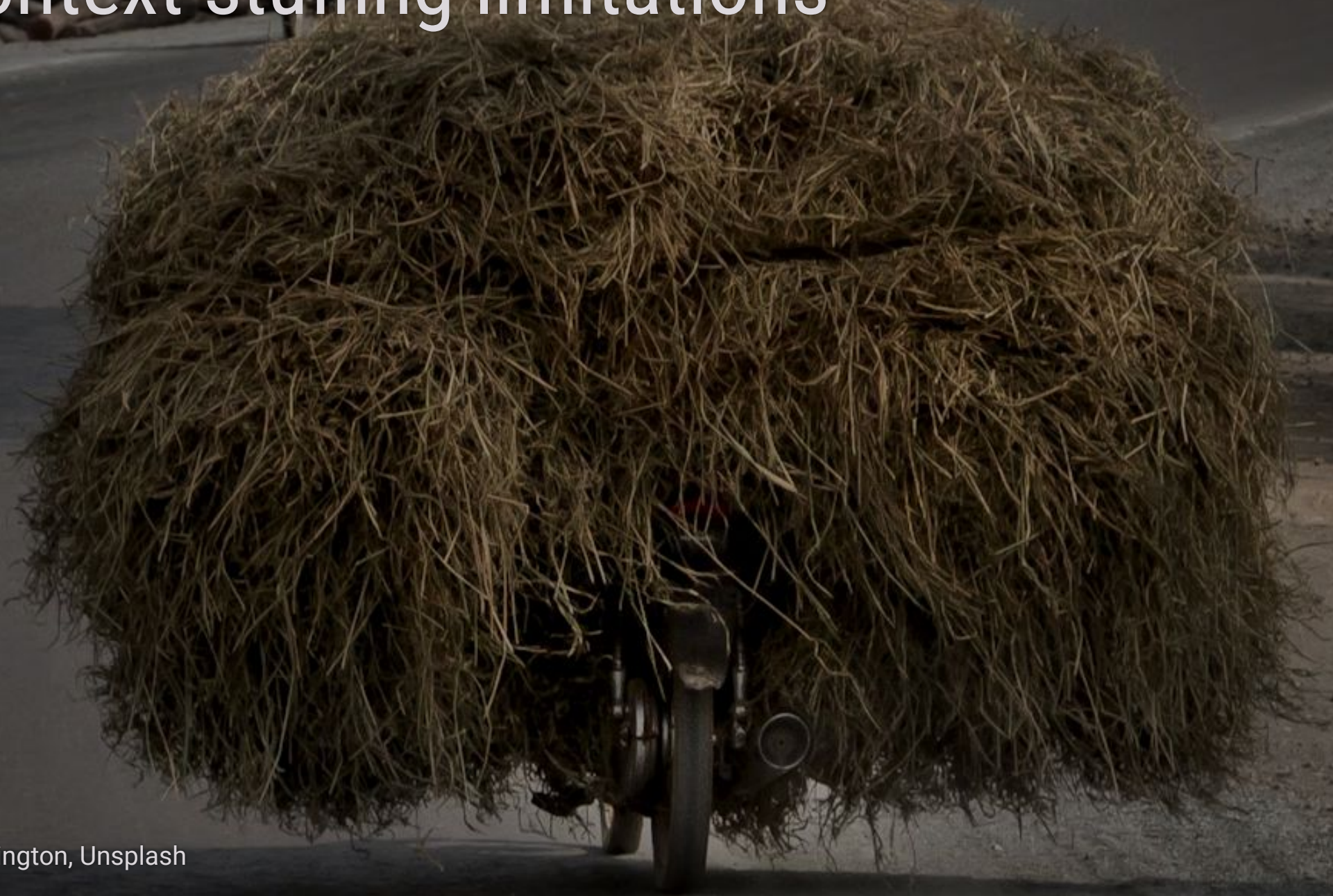
How do we test non-deterministic outputs in an automated manner?

LLM-as-a-judge?



A little bit about Retrieval Augmented Generation

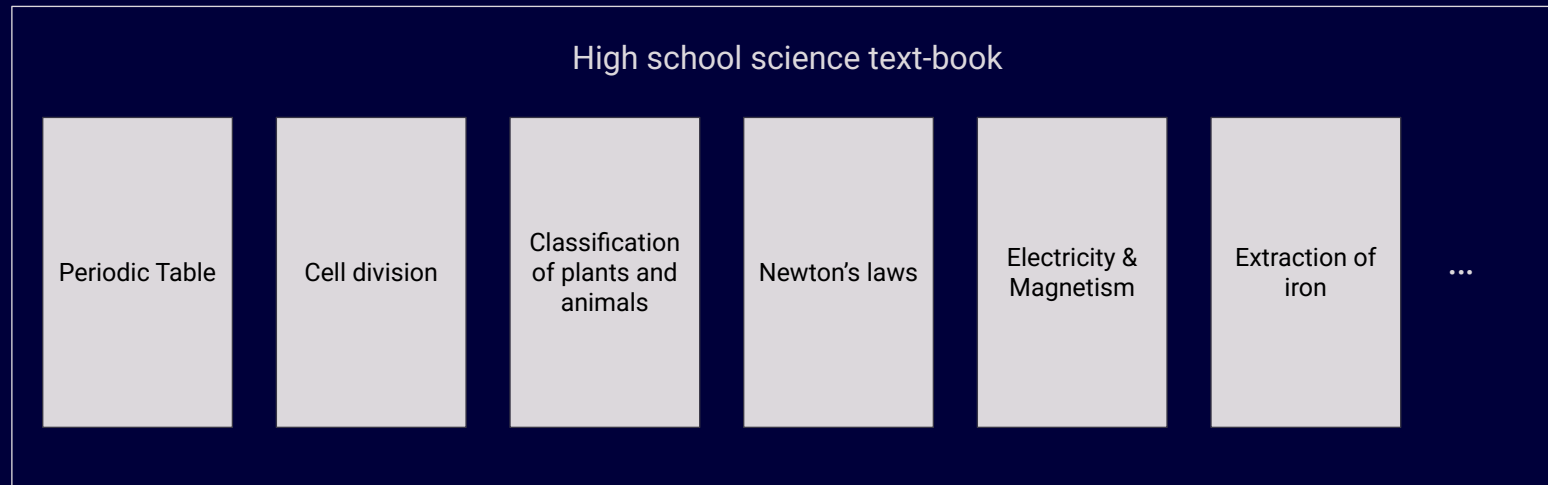
Context stuffing limitations



Credits: Ray Harrington, Unsplash

When we ask an LLM...

We upload a high-school science textbook to an LLM...

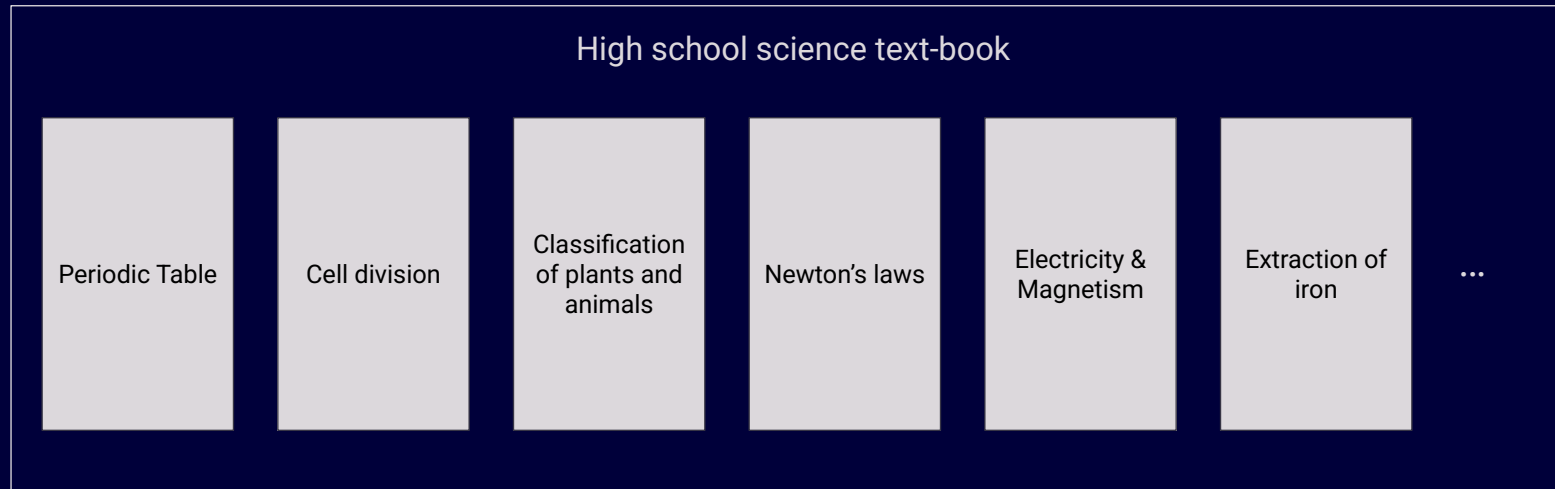


... and ask the LLM: Explain how a rocket works.

Context stuffed LLM

Explain how a rocket works.

LLM context window



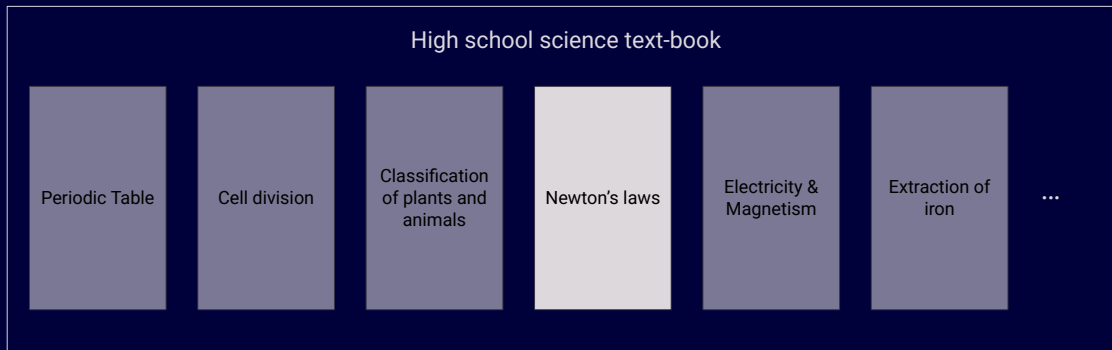
Generation

- Unreliable output
- Hallucination
- Higher costs

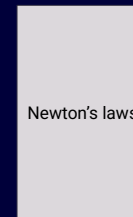
LLM with RAG

Explain how a rocket works.

Retrieval



LLM context window



Generation

- Targeted context
- Reliable output
- Lower costs

Terminology

Embeddings

Vector representations of text

E.g.

"How to reset my password?" $\rightarrow$ [0.11, -0.27, 0.53, ...]

"Change account password" $\rightarrow$ [0.12, -0.25, 0.50, ...]

Cosine similarity $\approx 0.98 \rightarrow$ semantically similar.

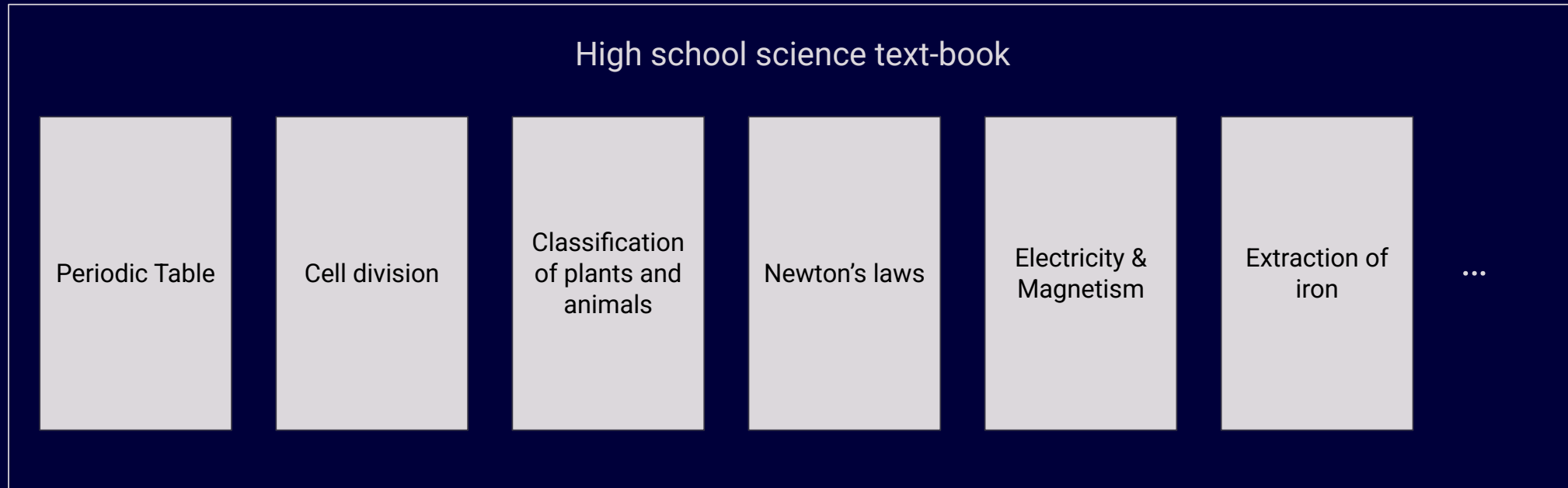
Vector Store / Vector Database

Database optimized for storing embeddings.

Terminology

Chunking

Breaking up document / data-source into smaller pieces



Demo outline

- ❖ Walk through RAG model setup
- ❖ Explore evaluations
- ❖ Debug evaluation with traces
- ❖ Demonstrate security issue

Data source

- ◆ [37 Signals Employee Handbook](#)

Show me. Don't tell me.



Where to use LLM-as-a-judge

- ❖ Smoke testing
- ❖ Prototypes

LLM-as-a-judge is no substitute for human evaluation

What will we need to do differently?

- ❖ Return to first principles
- ❖ Grapple with non-determinism
- ❖ Thinking statistically

Try out this demo!

The Beyond Quality community

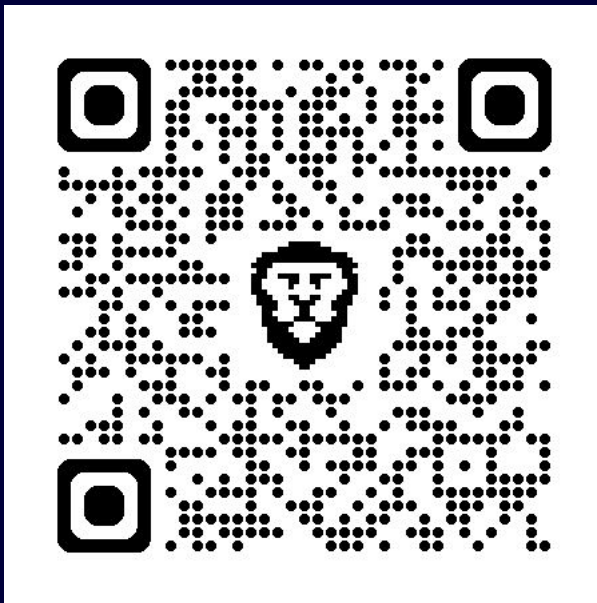
Beyond Quality website

- ❖ Pursue mastery collaboratively
- ❖ Translate ideas into research or talks
- ❖ Be part of a supportive community



Links

Source Repo



My LinkedIn

